Assignment - 3

Date
24/12/2022

1.(a) For the path $y = x$

$$\lim_{(x,y)\to(0,0)} \frac{3x^2|y|}{x^2+y^2} = \lim_{x\to 0} \frac{3x^2|x|}{x^2+x^2}$$

$$= \lim_{x\to 0} \frac{3|x|}{2}$$

$$= 0.$$

Claim: $\lim_{(x,y)\to(0,0)} \frac{3x^2|y|}{x^2+y^2} = 0$

$\longrightarrow$ Suppose $\left|\frac{3x^2|y|}{x^2+y^2} - 0\right| < \varepsilon$

whenever $\sqrt{x^2+y^2} < \delta$

We have,

$$\left|\frac{3x^2|y|}{x^2+y^2}\right| = 3|y| \cdot \left|\frac{x^2}{x^2+y^2}\right|$$

$$\leq 3|y|$$

$$\leq 3\sqrt{x^2+y^2}$$

$$< 3\delta$$

Thus, $\left|\frac{3x^2|y|}{x^2+y^2} - 0\right| < \varepsilon$

whenever $\sqrt{x^2+y^2} < \varepsilon/3$.

(b) Put $\theta = x+y+z$,

$$\lim_{(x,y,z)\to(0,0,0)} \frac{1-\cos(x+y+z)}{(x+y+z)^2} = \lim_{\theta\to 0} \frac{1-\cos\theta}{\theta^2}$$

$$= \lim_{\theta\to 0} \frac{2\sin^2\theta/2}{\theta^2}$$

$$= \frac{1}{2}\lim_{\theta\to 0}\left(\frac{\sin\theta/2}{\theta/2}\right)^2$$

$$= \frac{1}{2}\cdot 1^2 = \frac{1}{2}.$$

(c) $$\lim_{(x,y)\to(0,0)} \frac{\sin(x^2+y^2)}{x^2+y^2} = \lim_{\theta\to 0}\frac{\sin\theta}{\theta}$$

where $\theta = x^2+y^2$

$= 1.$

2. (a) Put $y = m_1 x$ & $z = m_2 x$

$$\lim_{(x,y,z)\to(0,0,0)} \frac{(x+y+z)^2}{x^2+y^2+z^2} = \lim_{x\to 0}\frac{(x+m_1x+m_2x)^2}{x^2+m_1^2x^2+m_2^2x^2}$$

$$= \frac{(1+m_1+m_2)^2}{1+m_1^2+m_2^2}$$

which depends upon m_1 & m_2

$\Rightarrow$ limit doesn't exist.

(b) Put $y = mx$

$$\cancel{\lim_{(x,y)\to(0,0)} mx^2 \cos(mx)}$$

$$\lim_{(x,y)\to(0,0)} \frac{xy\cos y}{x^2+y^2} = \lim_{x\to 0} \frac{mx^2\cos(mx)}{x^2+m^2x^2}$$

$$= \frac{m}{1+m^2}$$

$\Rightarrow$ limit doesn't exist.

3. $$f(x,y) = \begin{cases} \dfrac{\sin(xy)}{xy} & , xy \neq 0 \\ 1 & , xy = 0 \end{cases}$$

Put $xy = \theta$

$$\lim_{xy\to 0} f(x,y) = \lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1$$

$$= f(0,0)$$

$\Rightarrow f(x,y)$ is cont^n on $\mathbb{R}^2$.

4 $$f(x,y) = \begin{cases} \dfrac{x^3 y}{2x^4+y^2} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$$

For the path $y = x$

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{x\to 0} \frac{x^4}{2x^4+x^2} = \lim_{x\to 0} \frac{x^2}{2x^2+1}$$

$$= 0$$

<u>Claim:</u> $\lim_{(x,y)\to(0,0)} f(x,y) = 0.$

$\longrightarrow$ Suppose $|f(x,y) - 0| < \varepsilon$

whenever $\sqrt{x^2+y^2} < \delta$

We have,

$$|f(x,y) - 0| = \left| \frac{x^3 y}{2x^4+y^2} - 0 \right|$$

Put $a = \left| \frac{x^5}{2x^4+y^2} \right|$ & $b = \left| \frac{x y^2}{2x^4+y^2} \right|$

Since G. M. $\leq$ A. M.

$$\Rightarrow \sqrt{ab} \leq \frac{a+b}{2}$$

$$\Rightarrow \left| \frac{x^3 y}{2x^4+y^2} \right| \leq \frac{1}{2} \left| \frac{x^5 + x y^2}{2x^4+y^2} \right|$$

$$= \frac{1}{2} |x| \cdot \left| \frac{x^4+y^2}{2x^4+y^2} \right|$$

$$\leq \frac{1}{2} |x|$$

$$\leq \frac{1}{2} \sqrt{x^2+y^2} < \frac{1}{2} \delta$$

Thus, $|f(x,y) - 0| < \varepsilon$ whenever $\sqrt{x^2+y^2} < 2\varepsilon$

$\Rightarrow f(x,y)$ is cont^n at $(0,0)$.

5. $f(x,y) = e^{-\frac{1}{|x-y|}}$ when $x \neq y$

Clearly, $\lim_{|x-y| \to 0} f(x,y) = 0$

Thus, $f(x,y) = \begin{cases} e^{-\frac{1}{|x-y|}} & , x \neq y \\ 0 & , x = y \end{cases}$

is the required cont^n funct^n.

7. $f(x,y) = |x| + |y|$ & $g(x,y) = |xy|$

We have

$$f_x(0,0) = \lim_{x \to 0} \frac{f(x,0) - f(0,0)}{x - 0}$$

$$= \lim_{x \to 0} \frac{|x|}{x} \text{ doesn't exist}$$

$\Rightarrow f_x(0,0)$ doesn't exist,

Again, $$g_x(0,0) = \lim_{x \to 0} \frac{g(x,0) - g(0,0)}{x - 0}$$

$$= \lim_{x \to 0} \frac{0}{x} = 0$$

$\Rightarrow g_x(0,0)$ exists.

Now, suppose $x_0 \neq 0$

$$g_y(x_0, 0) = \lim_{y \to 0} \frac{g(x_0, y) - g(x_0, 0)}{y - 0}$$

$$= \lim_{y \to 0} \frac{|x_0 y| - 0}{y}$$

$$= |x_0| \lim_{y \to 0} \frac{|y|}{y} \text{ doesn't exist.}$$

$\Rightarrow g_y(x_0, 0)$ doesn't exist.

6. $$f(x, y) = \begin{cases} 1, & \text{when } x = 0 \text{ or } y = 0 \\ 0, & \text{otherwise} \end{cases}$$

Clearly, $f(x, y)$ is not cont^n at $(0, 0)$

(Suppose $\varepsilon < 1$ and $(x, y) \to (0, 0)$ but $x \neq 0, y \neq 0$
$|f(x, y) - f(0, 0)| = |1 - 0| \not< \varepsilon$)

Now,

$$f_x(0, 0) = \lim_{x \to 0} \frac{f(x, 0) - f(0, 0)}{x - 0}$$

$$= \lim_{x \to 0} \frac{1 - 1}{x} = 0.$$

Also $f_y(0, 0) = 0$.

Let $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$. ~~that~~ $f(x,y)$ is differentiable at (a,b). Then,

$$\lim_{(h,k)\to(0,0)} \phi(h,k) = 0$$

where $f(a+h, b+k) - f(a,b) = \left.\frac{\partial f}{\partial x}\right|_{(a,b)} h + \left.\frac{\partial f}{\partial y}\right|_{(a,b)} k + \sqrt{h^2+k^2}\,\phi(h,k)$.

8. (a). $f(x,y) = |x| + |y|$

We have

$$f_x(0,0) = \lim_{x\to 0} \frac{f(x,0) - f(0,0)}{x-0}$$

$$= \lim_{x\to 0} \frac{|x|}{x} \text{ doesn't exist}$$

$\Rightarrow f(x,y)$ is not differentiable at $(0,0)$.

(e) $f(x,y) = \begin{cases} \dfrac{x^2 y}{\sqrt{x^2+y^2}}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$

We have,

$$f_x(0,0) = \lim_{x\to 0} \frac{f(x,0) - f(0,0)}{x-0} = \lim_{x\to 0} \frac{0}{x} = 0$$

Now, $f_y(0,0) = 0$

Now,

Let $f(0+h, 0+k) - f(0,0) = \left.\frac{\partial f}{\partial x}\right|_{(0,0)} h + \left.\frac{\partial f}{\partial y}\right|_{(0,0)} k$

$+ \sqrt{h^2+k^2}\ \phi(h,k)$

$$\Rightarrow \frac{h^2 k}{\sqrt{h^2+k^2}} - 0 = 0 + 0 + \sqrt{h^2+k^2}\ \phi(h,k)$$

$$\Rightarrow \phi(h,k) = \cancel{\frac{h^2 k}{\sqrt{h^2+k^2}}} = \frac{h^2 k}{h^2+k^2}$$

$$\Rightarrow \lim_{(h,k)\to(0,0)} \phi(h,k) = \lim_{(h,k)\to 0} \frac{h^2 k}{h^2+k^2}$$

$$= \lim_{h\to 0} \frac{h^2 \cdot h}{h^2+h^2}$$

where $k = h$

$$= 0.$$

<u>Claim:</u> $\lim_{(h,k)\to(0,0)} \phi(h,k) = 0$

Suppose $|\phi(h,k) - 0| < \varepsilon$ whenever $\sqrt{h^2+k^2} < \delta$

where

$$\left|\frac{h^2 k}{h^2+k^2}\right| \le |k| \le \sqrt{h^2+k^2} < \delta$$

So, $|\phi(h,k) - 0| < \varepsilon$ whenever $\sqrt{h^2+k^2} < \varepsilon$.

9. $$f(x,y) = \begin{cases} (x-y)^2 \sin\frac{1}{x-y} & , x \neq y \\ 0 & , x = y \end{cases}$$

i) When $x \neq y$,

$$f_x(x,y) = 2(x-y)\sin\frac{1}{x-y} + \cancel{(x-y)^2} \times (-)\frac{1}{x-y} \times \frac{-1}{\cancel{(x-y)^2}}$$

$$= 2(x-y)\sin\frac{1}{x-y} - (-)\frac{1}{x-y}$$

$$f_y(x,y) = -2(x-y)\sin\frac{1}{x-y} + \cancel{(x-y)^2}(-)\frac{1}{(x-y)} \times \frac{1}{\cancel{(x-y)^2}}$$

$$= -2(x-y)\sin\frac{1}{(x-y)} + (-)\frac{1}{x-y}$$

ii) When $x = y = a$

$$f_x(a,a) = \lim_{x \to a} \frac{f(x,a) - f(a,a)}{x-a}$$

$$= \lim_{x \to a} \frac{(x-a)^{\cancel{2}} \sin\frac{1}{x-a} - 0}{\cancel{(x-a)}}$$

$$= \lim_{x \to a} (x-a) \cdot \sin\frac{1}{x-a}$$

$$= 0$$

Also, $f_y(a,a) = 0$

$\therefore f_x(x,x) = f_y(x,x) = 0.$

(a) $\therefore f_x$ & f_y exist $\forall (x,y) \in \mathbb{R}^2$

(b). Let $f(0+h, 0+k) - f(0,0) = \left.\frac{\partial f}{\partial x}\right|_{(0,0)} h$

$$+ \left.\frac{\partial f}{\partial y}\right|_{(0,0)} k + \sqrt{h^2+k^2}\,\phi(h,k)$$

$$\Rightarrow (h-k)^2 \sin\frac{1}{h-k} = 0 = 0 + 0 + \sqrt{h^2+k^2}\,\phi(h,k)$$

$$\Rightarrow \phi(h,k) = (h-k)^{3/2} \sin\frac{1}{h-k}$$

$$\Rightarrow \lim_{(h,k)\to(0,0)} \phi(h,k) = 0$$

$\Rightarrow$ f is differentiable at $(0,0)$.

(c). We have,

$$f_x(x,y) = \begin{cases} 2(x-y)\sin\frac{1}{x-y} - \cos\frac{1}{x-y}, & x \neq y \\ 0, & x = y \end{cases}$$

~~Clearly, $\lim_{(x,y)\to(0,0)} f_x(x,y)$ does~~

Clearly $\lim_{x\to y} f_x(x,y)$ doesn't exist

$\Rightarrow f_x$ is not cont^n on the line $y=x$.

10. Let $z = f(x, y)$, $x = r\cos\theta$ & $y = r\sin\theta$

(a). Use the chain rule,

$$\frac{\partial z}{\partial r} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial r}$$

$$= \frac{\partial f}{\partial x} \cos\theta + \frac{\partial f}{\partial y} \sin\theta$$

Now, $$\frac{\partial z}{\partial \theta} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial \theta}$$

$$= \frac{\partial f}{\partial x}(-r\sin\theta) + \frac{\partial f}{\partial y}(r\cos\theta)$$

$$\Rightarrow \frac{1}{r}\frac{\partial z}{\partial \theta} = -\frac{\partial f}{\partial x}\sin\theta + \frac{\partial f}{\partial y}\cos\theta.$$

(b) $f(x, y) = x^2 + 2xy$

$$\Rightarrow \frac{1}{r}\frac{\partial z}{\partial \theta} = -\frac{\partial f}{\partial x}\sin\theta + \frac{\partial f}{\partial y}\cos\theta$$

$$= -(2x + 2y)\sin\theta + 2x \cdot \cos\theta$$

$$\Rightarrow \frac{\partial z}{\partial \theta} = -(2x + 2y) \cdot r\sin\theta + 2x \cdot r\cos\theta$$

$$= -(2x + 2y)\,y + 2x \cdot x$$

$$= 2(x^2 - xy - y^2)$$